

Haoyan Luo

Ph.D. Student in Computer Science | ML Interpretability & Control
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Education

University of Cambridge

Ph.D. in Computer Science; Supervisor: Mateja Jamnik

Cambridge, UK
May 2025 - Present

- **Research topics:** Interpretability and Control of Large Language Models

Imperial College London

M.Res. in AI & Machine Learning; Supervisor: Lucia Specia

London, UK
Sep. 2023 - Sep. 2024

- **Thesis:** Explainability in Large Language Models: Pathways to Refinement and Alignment
- **Award:** Endowment Fellowship; Distinction Honours

The Chinese University of Hong Kong, Shenzhen

B.Eng. in Computer Science; First-Class Honours

Hong Kong / Shenzhen, China
Sep. 2019 - July 2023

- **Honors:** Dean's List of School of Data Science (AY21-23); Undergraduate Research Awards (AY21-23); Undergraduate Student Teaching Fellow Award (AY21-22); Shaw College Outstanding Graduates Award
- **Services:** President of TEDxCUHK(SZ); Student Coordinator, Shaw College

Experience

Vectify AI

Research Intern (Part-Time)

London, UK
April 2025 - Nov. 2025

- **PageIndex** : Built a reasoning-based RAG system ([link](#)) that simulates how human experts navigate and extract knowledge from long documents through tree search; Github 30K+ stars.

J.P. Morgan

AI & NLP Associate

London, UK
Sep. 2024 - March 2025

- **TIDE - Topic Insights and Document Exploration** : Led the development of topic modeling and topic distillation pipelines in TIDE, our internal state-of-the-art solution powered by LLMs.
- **Agentic Hallucination Detection** : Developed unified hallucination detection pipelines tailored for financial agents.

Shanghai AI Laboratory

Research Intern, supervised by Dr. Xun Zhao

Shanghai, China
July 2023 - Sep. 2023

- **Mechanistic LLM Interpretation**: Utilise the trained or customised LM head, discover that the hidden states of transformers can be viewed as large memory blocks, with each of its cells being highly decomposable and explainable;

Microsoft Research Asia

Research Intern, supervised by Dr. Jiang Bian & Dr. Wei Cao

Beijing, China
July 2022 - Nov. 2022

- **Feature engineering**: Co-proposed OpenFE, a novel framework with feature boosting method to accurately identify useful new features for tabular data. Github ([code](#)) 1K+ stars;

Deep Reinforcement Learning Group

Student Researcher, supervised by Prof. Shuang Li

Shenzhen, China
May 2022 - Nov. 2022

- **Meta-policy on temporal point process**: Researched multivariate temporal point process and formulate the traffic congestion and epidemic (COVID-19) curbing problems as finite-time horizon RL problems. Embed temporal point process dynamics into a meta-policy learning framework and extract explainable and transferrable information the trajectories ([code](#));

Experience (continued)

Tencent

Shenzhen, China

Big Data Engineer Intern

June 2021 – Sep. 2021

- **Recommendation system:** Built a recommendation system sub-module with a session-based recurrent neural network and deployed it within the "news" page and "Daily Q&A" page in WeChat Mini Program;
- **Data warehouse construction:** Built Hive and ClickHouse support and used Scala (together with SQL insertion and Spark & Hadoop embedded features) to query information from the warehouse for data analysis;

Seasun Entertainment, Kingsoft

Zhuhai, China

Software Engineer Intern

June 2020 – August 2020

- **Backend development:** Built and maintained a large-scale intranet used to monitor users' behaviors and dialogue contents in a popular web game in China, JX3, with over 5 million active users;

Publications

Conference / Workshop Papers

1. **Haoyan Luo**, Mateo Espinosa Zarlenga, Mateja Jamnik. **Don't Lose Focus: Activation Steering via Key-Orthogonal Projections**, *ICML'26 Mechanistic Interpretability*.
2. **Haoyan Luo**, Lucia Specia. **Tuning Language Models by Mixture-of-Depths Ensemble**, *COLM'26 Actionable Interpretability*.
3. Tianping Zhang, Zheyu Zhang, Zhiyuan Fan, **Haoyan Luo**, Fengyuan Liu, Wei Cao, Jian Li. **OpenFE: Automated Feature Generation with Expert-level Performance**, *ICML'23*.

Preprints

1. **Haoyan Luo**, Mateo Espinosa Zarlenga, Mateja Jamnik. **Persistent Sparse Autoencoders: Learning Feature-Specific Timescales in Language Model Representations**, *arXiv:2607.17117*, 2026.
2. Sae Young Moon, Myeong-jun Erik Jang, **Haoyan Luo**, Chunyang Xiao, Antonios Georgiadis, Fran Silavong. **TIDE - Industry-aligned Granular Topic Modeling**, *arXiv:2601.11762*, 2026.
3. **Haoyan Luo**, Lucia Specia. **From Understanding to Utilization: A Survey on Explainability for Large Language Models**, *arXiv:2401.12874*, 2024.

Competitions and Awards

- Endowment Fellowship, Imperial College London 2023
- Meritorious Winner, Mathematical Contest in Modeling (MCM'22) 2022
- First Prize in Contemporary Undergraduate Mathematical Contest in Modeling, China (CUMCM'21) 2021
- Third Prize, RoboMaster University AI Challenge 2021
- National Silver Award, First Prize Award in Guangdong, Google App Inventor Challenge 2019

Service & Teaching

- **Reviewer:** EMNLP'24, '25, '26; ICML'25, '26; ICLR'25.
- **Teaching Assistant:** CUHK: CSC3170 (*Database Systems*) and DDA4020 (*Advanced Machine Learning*); University of Cambridge: *Machine Learning for Real-World Data*.

Technical Skills

Programming: Python, C++, PyTorch, Rust, MATLAB, Scala, Matlab

Languages: Mandarin Chinese (Native), Cantonese (Fluent), English (Professional)